

# Preventable Hospitalisations Among Indigenous Australians: A Regional and Longitudinal Analysis



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## Executive Summary

Potentially preventable hospitalisations (PPH) are admissions for conditions where timely access to quality community-based care could have avoided hospital treatment. Indigenous Australians are hospitalized for preventable conditions at rates far exceeding those of non-Indigenous Australians, a persistent gap rooted in inequitable access to primary healthcare, chronic disease management, and the social determinants of health.

An OLS regression on all 31 Primary Health Networks (PHNs) showed that geographic remoteness was the strongest predictor of regional PPH rates (coefficient = 2.82 admissions per 100 residents per remoteness step,  $p = 0.011$ ), accounting for 73.9% of variance alongside SEIFA disadvantage rank and health check uptake. A Random Forest model predicted average length of stay with high accuracy (Test  $R^2 = 0.95$ , Test MAE = 0.34 days), with PPH condition type explaining approximately 90% of predictive variance. The top three reasons for hospital admissions were cellulitis, COPD, convulsions and epilepsy. All top ten conditions fell within the chronic or acute PPH categories, none were vaccine preventable

The results show that improving access to healthcare matters more than increasing vaccination rates in reducing preventable hospital visits among Indigenous Australians.

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## **Section 2: Introduction**

Potentially preventable hospitalisations (PPH) reflect primary care failures. For Indigenous Australians, PPH rates are two to five times higher than non-Indigenous Australians, highlighting severe system inequity [1]. Using six years of AIHW data, this project combines regression and machine learning to analyze regional drivers and predict future clinical severity.

## **Section 3: Project Aims and Objectives**

The aim of this project was to identify structural drivers of preventable hospitalisation burden among Indigenous Australians and to build a robust predictive model for clinical severity at the condition level. To achieve this aim, we asked the following research questions:

RQ1: Which specific conditions account for the greatest volume of PPH admissions for Indigenous Australians across 2012–2018, and do they cluster in chronic, acute, or vaccine-preventable categories?

RQ2: Does geographic remoteness or socioeconomic disadvantage more strongly predict regional PHN-level PPH rates, after controlling for health check investment?

RQ3: Can machine learning accurately predict average length of stay from condition-level features, and which features drive that prediction?

## **Section 4: Methodology**

### **4.1 Overview**

We used two AIHW datasets covering Indigenous Australians across Australian Primary Health Networks. After cleaning and restructuring both datasets, we carried out EDA to describe condition level and regional patterns. We then built an OLS regression model to explain regional PPH rates, and four machine learning models to predict average length of stay at the condition level. All analysis was conducted in Python.

### **4.2 Datasets**

Two datasets were obtained from the Australian Institute of Health and Welfare (AIHW) [2] and used in this analysis (Table 1)

| Dataset                                | Source     | Coverage                                                 | Key Variables                                                                                                       |
|----------------------------------------|------------|----------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|
| A1: PHN-Level Master Dataset           | AIHW, 2020 | 31 PHNs, cross-sectional                                 | PPH rate per 100 residents, health checks per 1,000 Indigenous residents, remoteness classification, SEIFA quintile |
| B1: Indigenous Condition-Level Dataset | AIHW, 2020 | 2012–13 to 2017–18, condition $\times$ sex $\times$ year | PPH admissions, bed days, average length of stay, PPH category, PPH condition, sex, year                            |

*Table 1. Summary of the two AIHW datasets used in this analysis. PHN = Primary Health Network; PPH = potentially preventable hospitalisation; SEIFA = Socio-Economic Indexes for Areas.*

We excluded derived columns to prevent circularity, ordinal-encoded remoteness, and treated SEIFA quintiles [4] as numeric ranks. No imputation was needed for the 31 complete PHNs. For the condition dataset, we removed subtotal rows to avoid double counting. We also excluded 76 rows with suppressed missing values instead of imputing them, following established privacy protocols for small counts [2].

### 4.3 Condition-Level Burden (RQ1)

Analysis of total PPH admissions across the 2012–2018 period (Figure 1) showed cellulitis as the leading condition by admission volume, followed by COPD and convulsions and epilepsy. All top 10 conditions fell into the chronic or acute PPH categories. No vaccine preventable condition appeared in the top 10, indicating that the primary gap is in community management of chronic disease and timely response to acute presentations, rather than immunisation coverage.

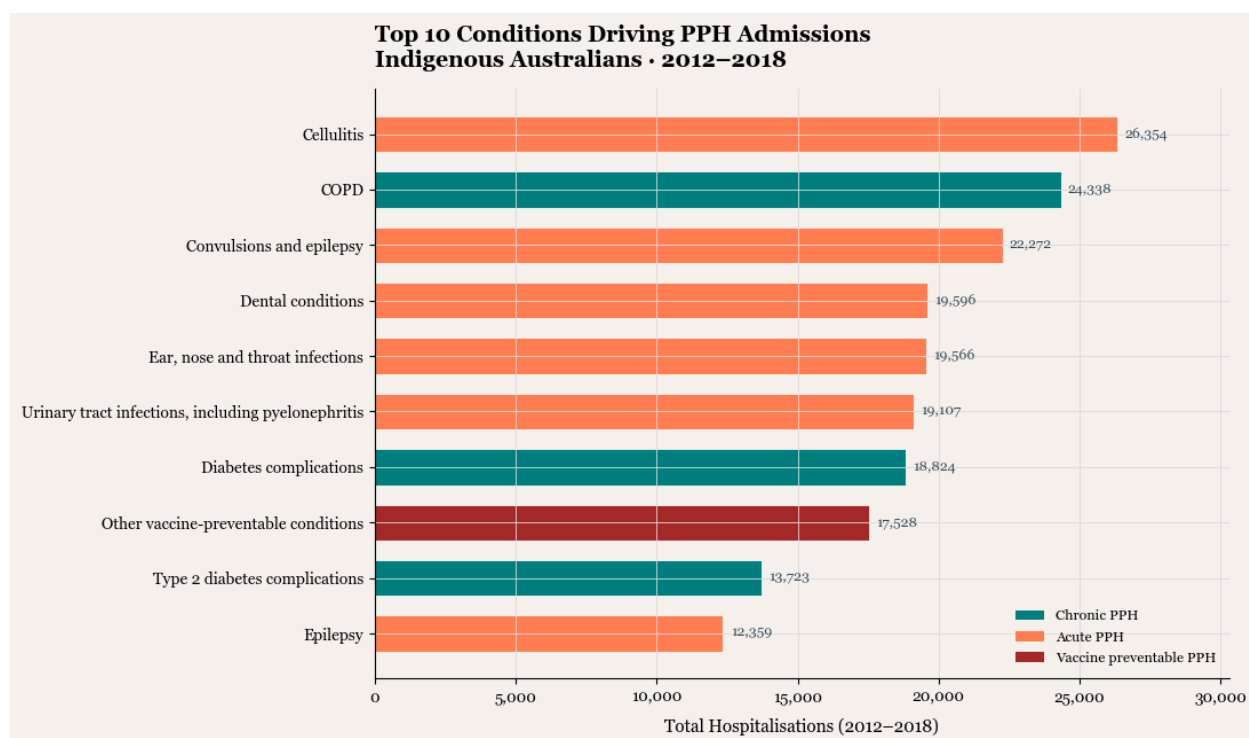


Figure 1: Bar chart of top 10 PPH conditions by total admissions, 2012–2018.

#### 4.4 Regional Variation (RQ2)

PHN-level PPH rates ranged from approximately 1.4 per 100 residents in major city PHNs to 29.4 in the Northern Territory (the sole Very Remote PHN). A systematic remoteness gradient was visible across all five remoteness categories, though within category variation particularly in Inner Regional PHNs, suggests that remoteness alone does not fully determine regional performance (Figure 2).

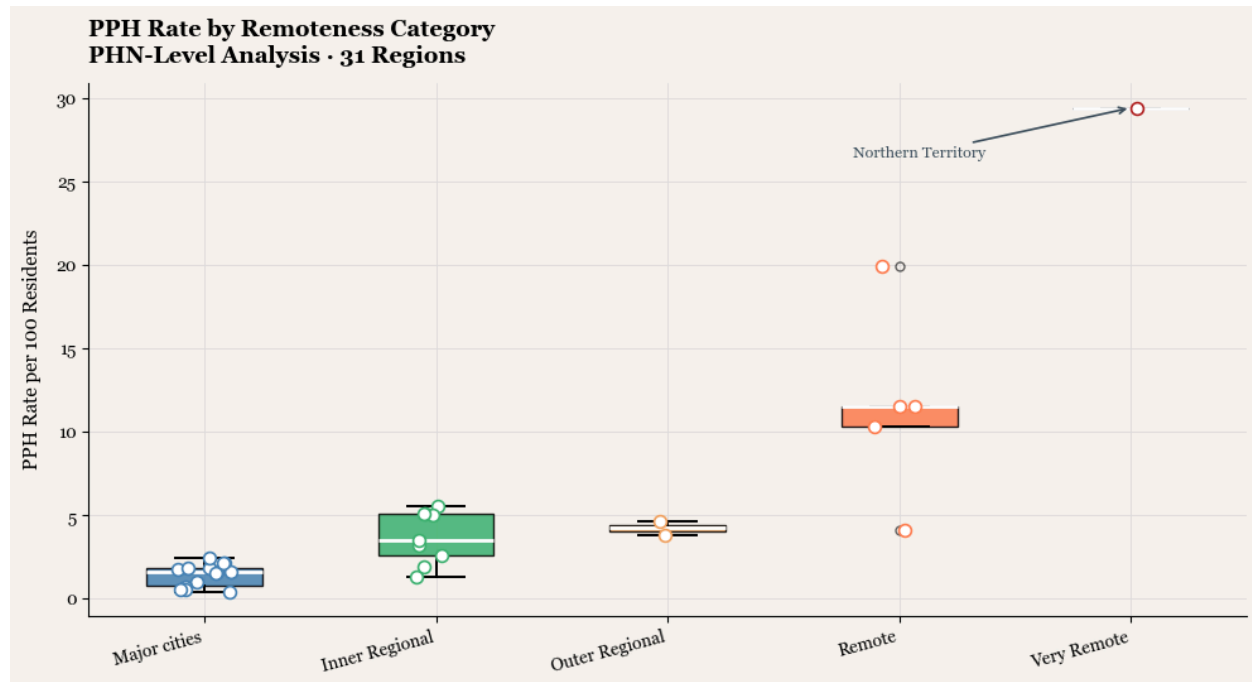


Figure 2: Box plot of PPH rate per 100 residents by remoteness category.

## 4.5 Statistical Models

**OLS Regression (RQ2):** To answer RQ2, we fitted an ordinary least squares regression to the 31 PHN dataset. The dependent variable was the PPH rate, with remoteness rank, SEIFA rank, and health check uptake as predictors. We checked VIF for multicollinearity and used diagnostic plots in (Figure 3) and (Figure 4) to evaluate model assumptions. The Northern Territory PHN was flagged as a high leverage point, so we used Cook's Distance to assess its influence.

**Machine Learning Models (RQ3):** To answer RQ3, we trained Random Forest and Gradient Boosting models, using Linear and Ridge Regression as baselines. Features included reporting year, sex, category, and condition, using label encoding. We excluded aggregate 'Persons' data to prevent leakage. The data was split 70/15/15 into training, validation, and test sets. We evaluated performance using  $R^2$ , MAE, and RMSE, and extracted feature importance from the Random Forest.

## Section 5: Results

### 5.1 OLS Regression: Regional PPH Rate Predictors (RQ2)

#### 5.1.1 OLS (Model Performance)

The OLS model explained 73.9% of the variance in regional PPH rates (Adjusted  $R^2 = 0.709$ ,  $p\text{-value} < 0.001$ ). Remoteness rank was the strongest predictor, with each one step increase adding 2.82 PPH admissions per 100 residents ( $p\text{-value} = 0.011$ ). Health check uptake was also positively associated ( $p = 0.018$ ), reflecting concentration of preventive investment in high burden communities. SEIFA rank was not statistically significant ( $p = 0.507$ ).

### 5.1.2 Assumptions & Diagnostics

Model assumptions were broadly satisfied. However, the Northern Territory PHN was an outlier, with Cook's Distance confirming it as a high leverage point. VIF analysis showed multicollinearity between remoteness rank (VIF = 13.1) and health check uptake (VIF = 17.7). Both variables were retained with this condition noted, so individual coefficients should be interpreted with caution.

| Predictor                                    | Coefficient | Std. Error | t-statistic | p-value |
|----------------------------------------------|-------------|------------|-------------|---------|
| (Intercept)                                  | -2.94       | 2.17       | -1.36       | 0.186   |
| Remoteness Rank                              | 2.82*       | 1.03       | 2.74        | 0.011   |
| SEIFA Rank                                   | 0.49        | 0.73       | 0.67        | 0.507   |
| Health Checks per 1,000 Indigenous Residents | 0.04*       | 0.02       | 2.50        | 0.018   |

Table 2. OLS regression results. Dependent variable: PPH rate per 100 residents.

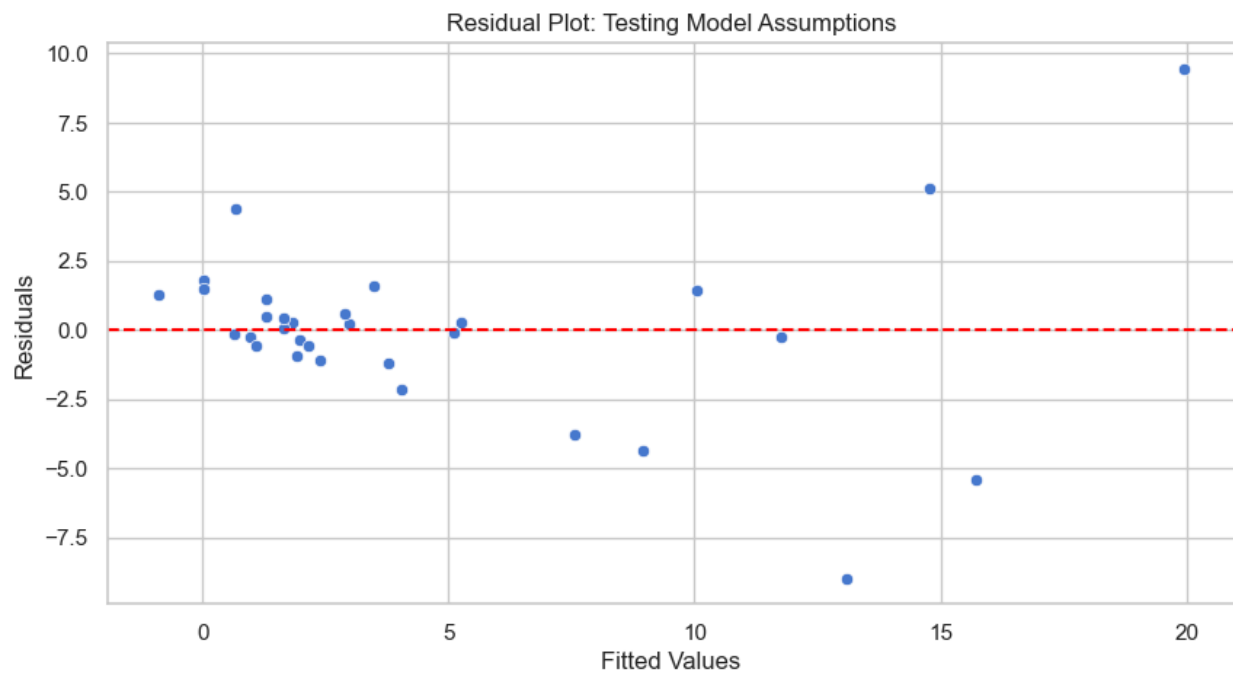


Figure 3. Residual Plot



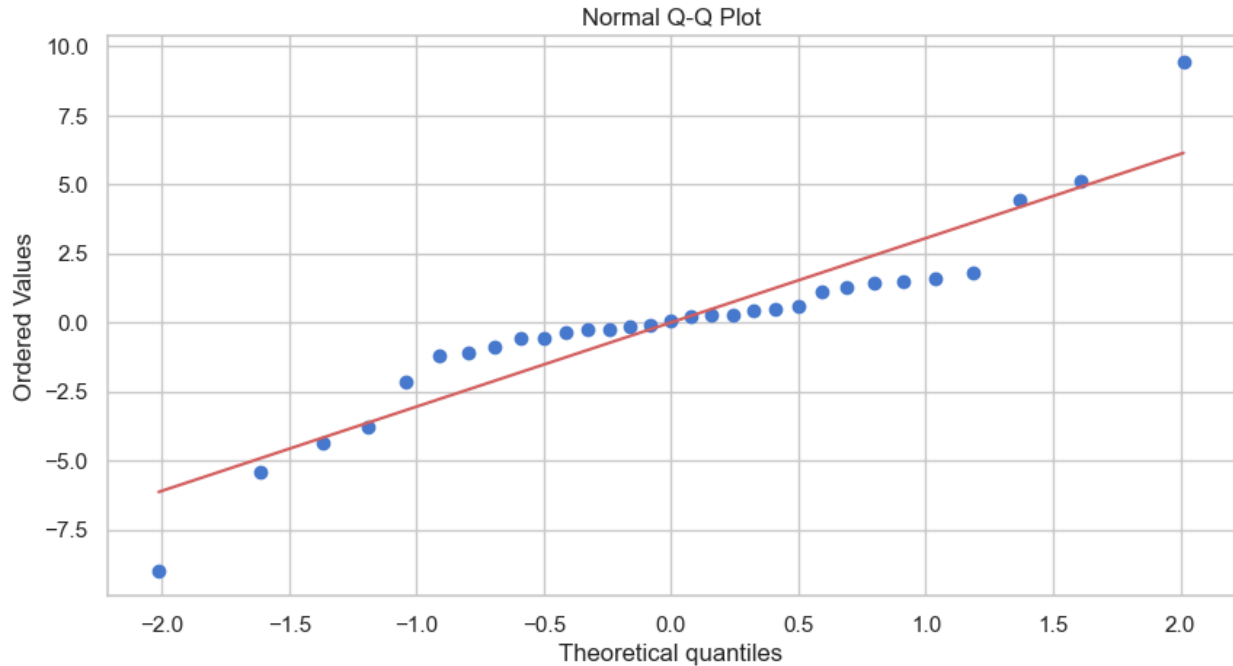


Figure 4. Q-Q Plot

The residual plot (Figure 3) showed no systematic pattern, mostly supporting linearity and homoscedasticity, though one high leverage point is visible at higher fitted values consistent with the NT PHN. The Q-Q plot (Figure 4) indicated approximate normality in the central range, with deviation at the tails driven by the same outlier.

### 5.1.3 Random Forest: Predicting Average Length of Stay (RQ3)

Table 3 shows model performance on the test set. Linear and Ridge regression performed poorly (Test  $R^2$  approximately 0.11 and 0.10), confirming the relationship is highly non-linear. The Random Forest model performed best (Test  $R^2 = 0.952$ , Test MAE approximately 0.34 days), with Gradient Boosting following closely behind. While the Random Forest showed mild overfitting (Train  $R^2 = 0.987$  vs. Val  $R^2 = 0.904$ ), the high test score (0.952) confirms it generalises well to unseen data.

| Model                               | Train $R^2$  | Val $R^2$    | Test $R^2$   | Test MAE (days) | Test RMSE (days) |
|-------------------------------------|--------------|--------------|--------------|-----------------|------------------|
| Linear Regression                   | 0.12         | 0.10         | 0.11         | 1.45            | 2.18             |
| Ridge Regression ( $\alpha = 1.0$ ) | 0.12         | 0.10         | 0.10         | 1.46            | 2.19             |
| Gradient Boosting                   | 0.971        | 0.882        | 0.931        | 0.51            | 0.76             |
| <b>Random Forest (best)</b>         | <b>0.987</b> | <b>0.904</b> | <b>0.952</b> | <b>0.34</b>     | <b>0.63</b>      |

Table 3. Model performance comparison for predicting average length of stay (days).

Figure 5 — Random Forest: Predicted vs Actual & Residuals (Test Set)

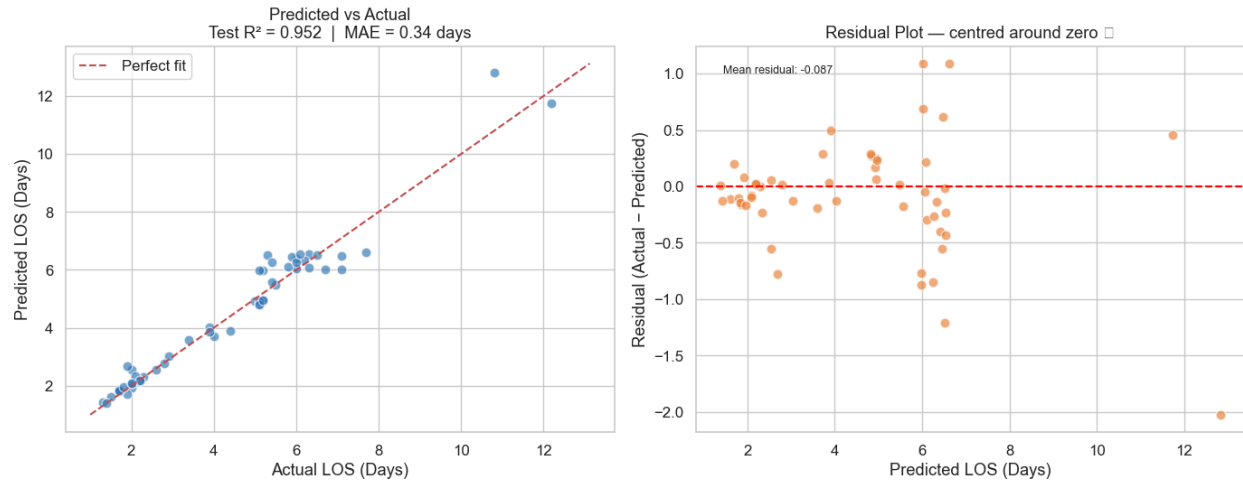


Figure 5: Diagnostic plots for Random Forest

#### 5.1.4 Random Forest: Feature Importance

Feature importance analysis showed that PPH condition accounted for approximately 90% of variance explained, followed by PPH category at approximately 7%, with reporting year and sex contributing approximately 3% combined. This finding has direct clinical meaning. The specific disease a patient is admitted for is by far the strongest determinant of how long they will stay in hospital, and demographic or temporal features add relatively little information once the condition is known.

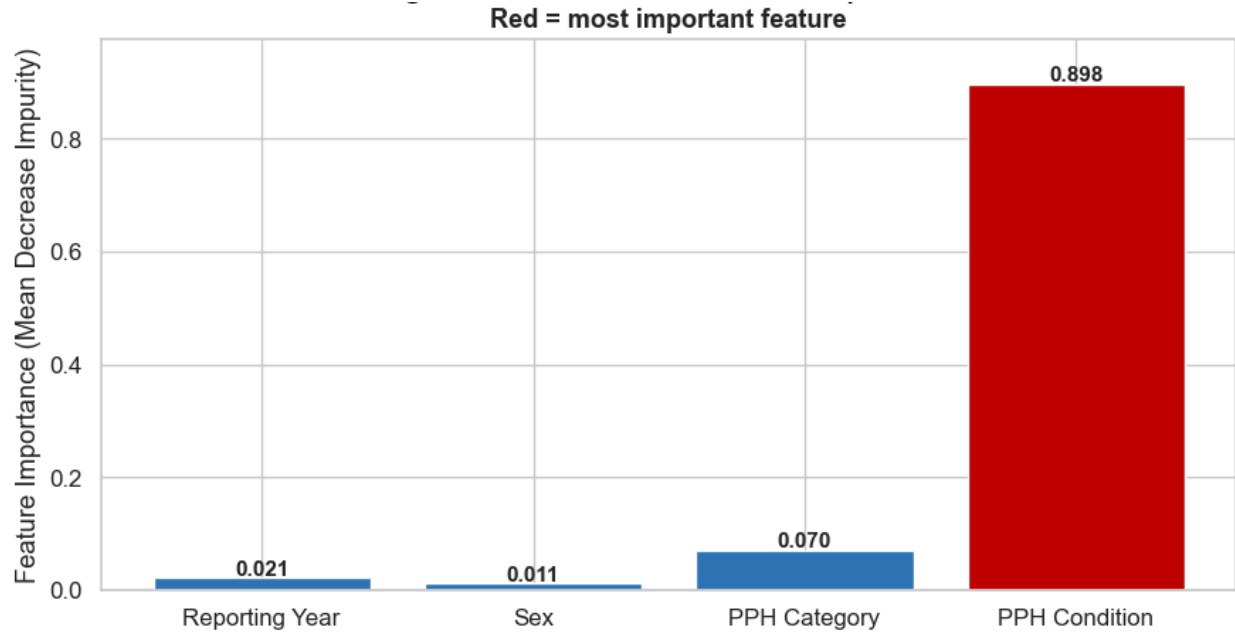


Figure 6: Feature importance bar chart from Random Forest.

## Section 6: Conclusion

### 6.1 Ethical Considerations

Working with Indigenous health data requires responsibilities that go beyond standard research ethics. Indigenous Australians have a right to self-determination over data describing their lives, a core principle of Indigenous Data Sovereignty and the CARE Principles [3].

To respect this, we excluded AIHW privacy-suppressed small cell counts rather than imputing them, protecting communities from identification. Crucially, high PPH rates must be framed as a systemic failure to deliver accessible, culturally safe primary care rather than a community deficit, as highlighted in national health performance frameworks [1].

Reflecting on Indigenous Governance and Accountability (IGA), this project highlights three key principles for professional data science:

- Research questions must be community informed and co-designed, not imposed externally.
- Predictive models can either help resource planning or harm communities via rationing, depending on how they are deployed.
- Results must be made fully accessible to the affected communities, not just kept in academic reports.

## 6.2 Limitations

The "Very Remote" category relies entirely on a single, high leverage data point (the Northern Territory PHN). Analyzing data at the broad PHN level also hides massive variations and inequities within those regions. Additionally, 76 privacy suppressed rows were excluded, meaning the machine learning model's accuracy may be lower for rare conditions.

## 6.3 Stakeholders

Policymakers can use the OLS evidence to support place-based, remote-first funding for primary care infrastructure, addressing the persistent shortages in rural healthcare access [5]. Meanwhile, hospital planners can use the Random Forest model to anticipate bed demand based on specific conditions. Future work should use post 2018 data, account for regional variations, and actively involve community representatives.

## References

- [1] Australian Institute of Health and Welfare (AIHW), "Aboriginal and Torres Strait Islander Health Performance Framework 2022 summary report," Cat. no. IHW 261, 2022. [Online]. Available: <https://www.aihw.gov.au/reports/indigenous-health-welfare/hpf-2022-summary>
  
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